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POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
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Salesforce Project Documentation

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Project Overview

I. Introduction

WhatNext Vision Motors is an automotive innovator focused on transforming the customer experience through technology-enabled mobility solutions. The Salesforce CRM project was designed to smoothen the customer ordering process, enhance operational efficiency, and ensure seamless communication among customers, sales teams, and dealer locations. The system promotes accuracy, transparency, and ease throughout the customer's journey through automated dealer suggestions, real-time stock validations, and scheduled order status processing.

II. Objectives

The key focus of this Salesforce implementation is to make ordering seamless and efficient for customers at WhatNext Vision Motors. Some of the things the system is supposed to achieve include

- Automate dealer location recommendations based on a customer's address.
- Prevent orders of vehicles not in stock from being placed.
- Schedule automatic updates to ensure accuracy in order status in real time.
- Improve internal operations by minimizing manual work and automating approval and ordering processes. Enhance customer satisfaction through clear communication and timeliness of updates.

PHASE 1: Requirement Analysis & Planning

→Understanding Business Requirements

- The company lacks a unified system to manage customers, vehicles, dealers, and orders.
- Stock availability is not monitored properly, causing invalid or overbooked vehicle orders.
- Dealer assignment is done manually, resulting in mismatches and delays.
- Test-drive reminders and service follow-ups are inconsistently carried out.
- Management needs dashboards to monitor sales performance and service activities.

→Defining Project Scope & Objectives

- Build custom objects for Vehicles, Dealers, and Orders.
- Implement nearest dealer identification using customer address.
- Automate stock validation prior to order creation.
- Introduce scheduled automation for updating order statuses.
- Ensure user profiles, security controls, and permissions match operational roles.

→Data Model & Security Model

- Objects: Customer, Dealer, Vehicle, Order.
- Relationships: Lookup relationships for Dealer → Order, Vehicle → Order.

→Stakeholder Mapping

- CRM Admin – Handles configurations and maintenance
- Sales Team – Creates orders and monitors availability
- Dealer Managers – Maintain stock and verify vehicle availability
- Customers – End users of the ordering process

→Execution Roadmap

1. Week 1: Requirement validation, object creation
2. Week 2: Flow development for dealer assignment
3. Week 3: Stock validation setup and automation
4. Week 4: Scheduled process build and testing
5. Week 5: Deployment and documentation

PHASE 2: SALESFORCE DEVELOPMENT – BACKEND & CONFIGURATIONS

→ Environment Setup & DevOps Workflow

Scratch orgs and sandboxes were used to configure features. Changes were version-controlled using Change Sets.

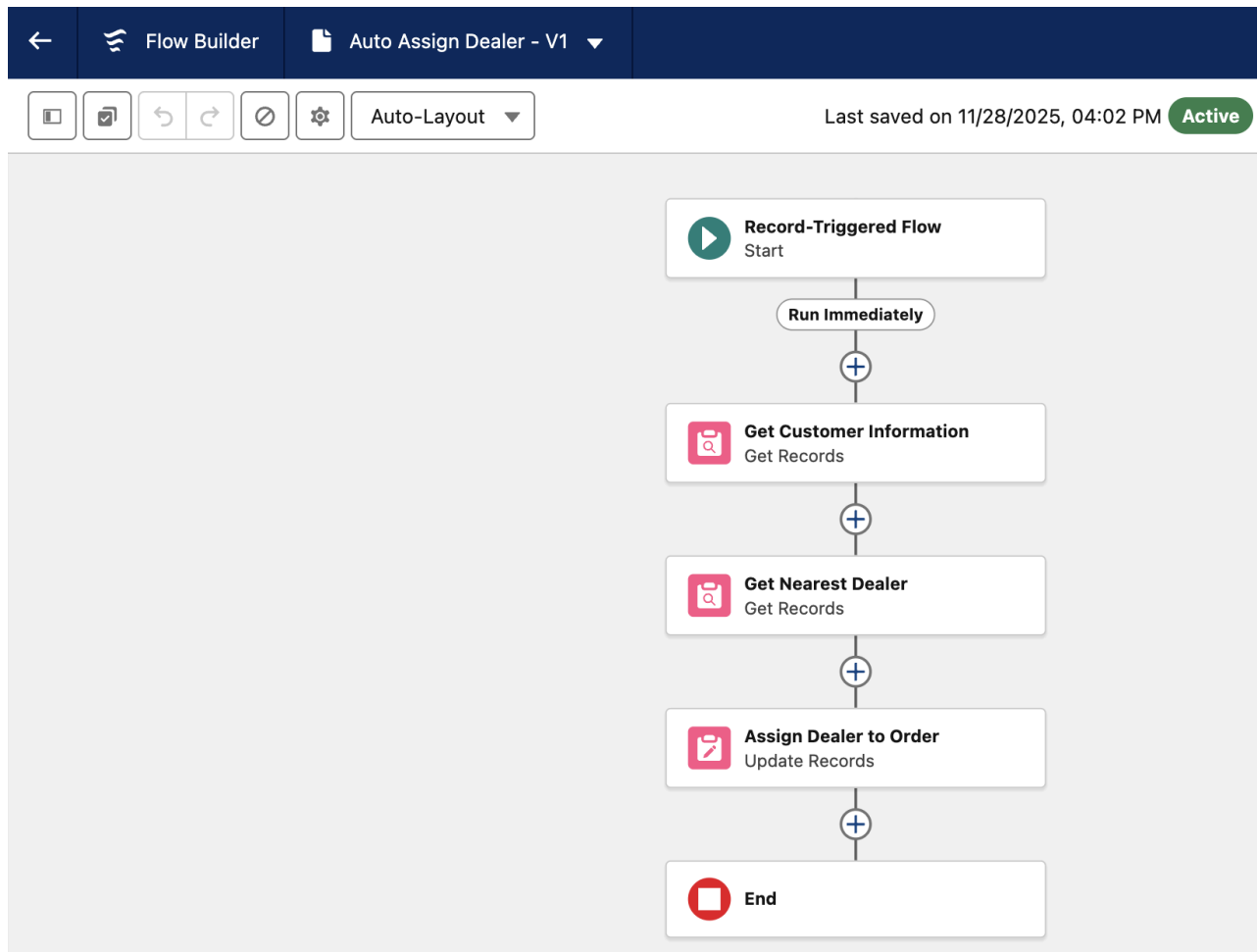
→Custom Objects, Fields, Validation Rules, Automation

- Vehicle Object: Includes fields for Model, Stock Quantity, Price.
- Order Object: Includes Customer, Dealer, Vehicle, Order Status.
- Validation Rule: Prevent order creation when Stock Quantity = 0.

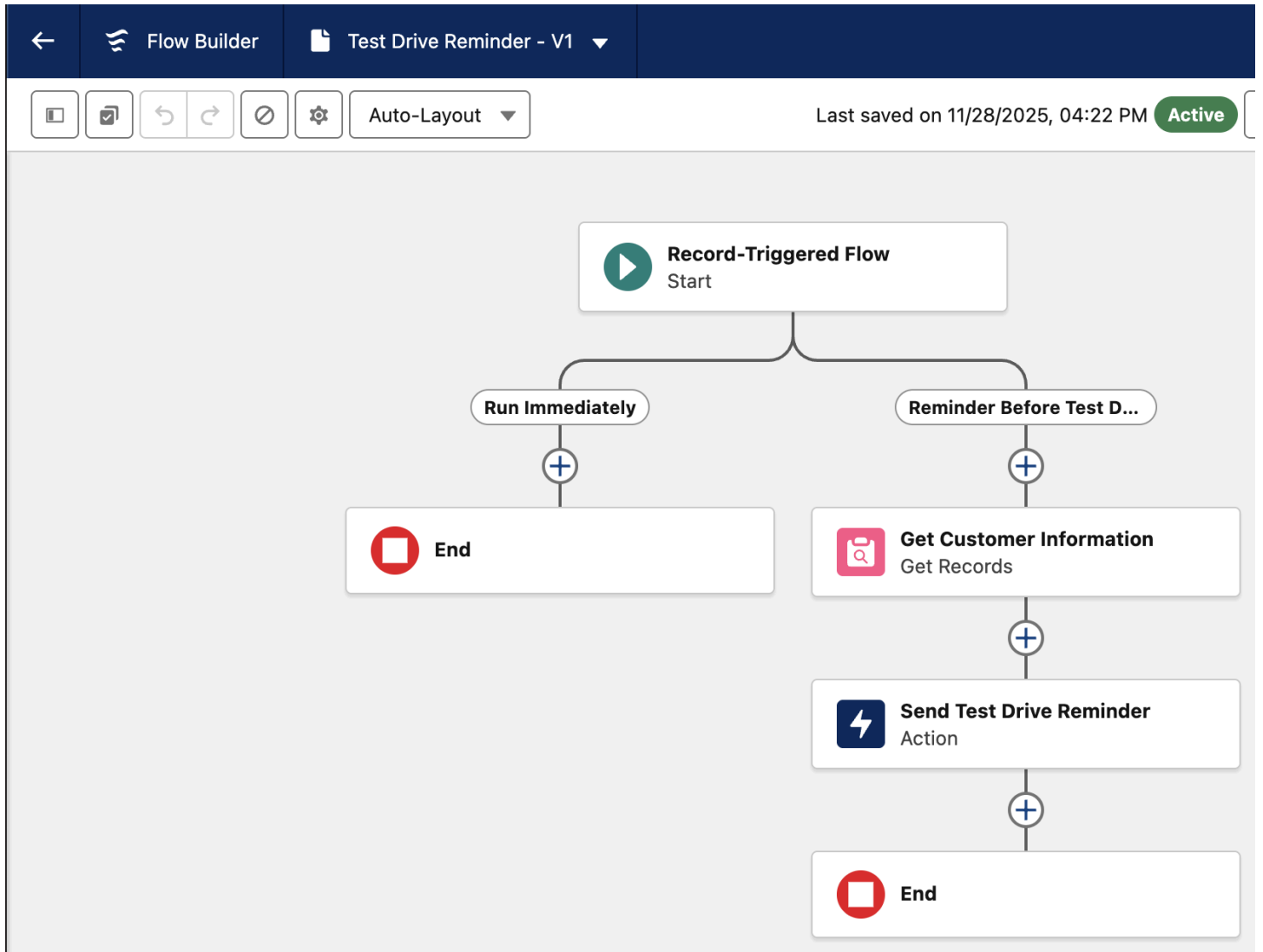
[Insert Screenshot: Validation Rule Screen]

→Automation

- Flow: Auto Assign Dealer



- Flow: Test Drive Reminder



→ Apex Classes/Triggers (If Any)

VehicleOrderTriggerHandler

File • Edit • Debug • Test • Workspace • Help • < • >

VehicleOrderTriggerHandler.apxc • VehicleOrderTrigger.apxt • VehicleOrderBatch.apxc • VehicleOrderBatchScheduler.apxc

Code Coverage: None • API Version: 65

```
1 public class VehicleOrderTriggerHandler {
2
3     public static void handleTrigger(List<Vehicle_Order__c> newOrders, Map<Id, Vehicle_Order__c> oldOrders, Boolean isBefore, Boolean isAfter, Boolean isInsert, Boolean isUpdate) {
4         if (isBefore && (isInsert || isUpdate)) {
5             preventOrderIfOutOfStock(newOrders);
6         }
7
8         if (isAfter && (isInsert || isUpdate)) {
9             updateStockOnOrderPlacement(newOrders);
10        }
11    }
12
13    private static void preventOrderIfOutOfStock(List<Vehicle_Order__c> orders) {
14        Set<Id> vehicleIds = new Set<Id>();
15        for (Vehicle_Order__c order : orders) {
16            if (order.Vehicle__c != null) {
17                vehicleIds.add(order.Vehicle__c);
18            }
19        }
20
21        if (!vehicleIds.isEmpty()) {
22            Map<Id, Vehicle__c> vehicleStockMap = new Map<Id, Vehicle__c>{
23                [SELECT Id, Stock_Quantity__c FROM Vehicle__c WHERE Id IN :vehicleIds]
24            };
25
26            for (Vehicle_Order__c order : orders) {
27                Vehicle__c vehicle = vehicleStockMap.get(order.Vehicle__c);
28                if (vehicle != null && vehicle.Stock_Quantity__c <= 0) {
29                    order.addError('This vehicle is out of stock. Order cannot be placed.');

Logs Tests Checkpoints Query Editor View State Progress Problems



| User | Application | Operation | Time | Status | Read |
|------|-------------|-----------|------|--------|------|
|------|-------------|-----------|------|--------|------|


```

VehicleOrderTrigger

File • Edit • Debug • Test • Workspace • Help • < • >

VehicleOrderTriggerHandler.apxc • VehicleOrderTrigger.apxt • VehicleOrderBatch.apxc • VehicleOrderBatchScheduler.apxc

Code Coverage: None • API Version: 65

```
1 trigger VehicleOrderTrigger on Vehicle_Order__c (before insert, before update, after insert, after update) {
2     VehicleOrderTriggerHandler.handleTrigger(trigger.new, trigger.oldMap, trigger.isBefore, trigger.isAfter, trigger.isInsert, trigger.isUpdate);
3 }
```

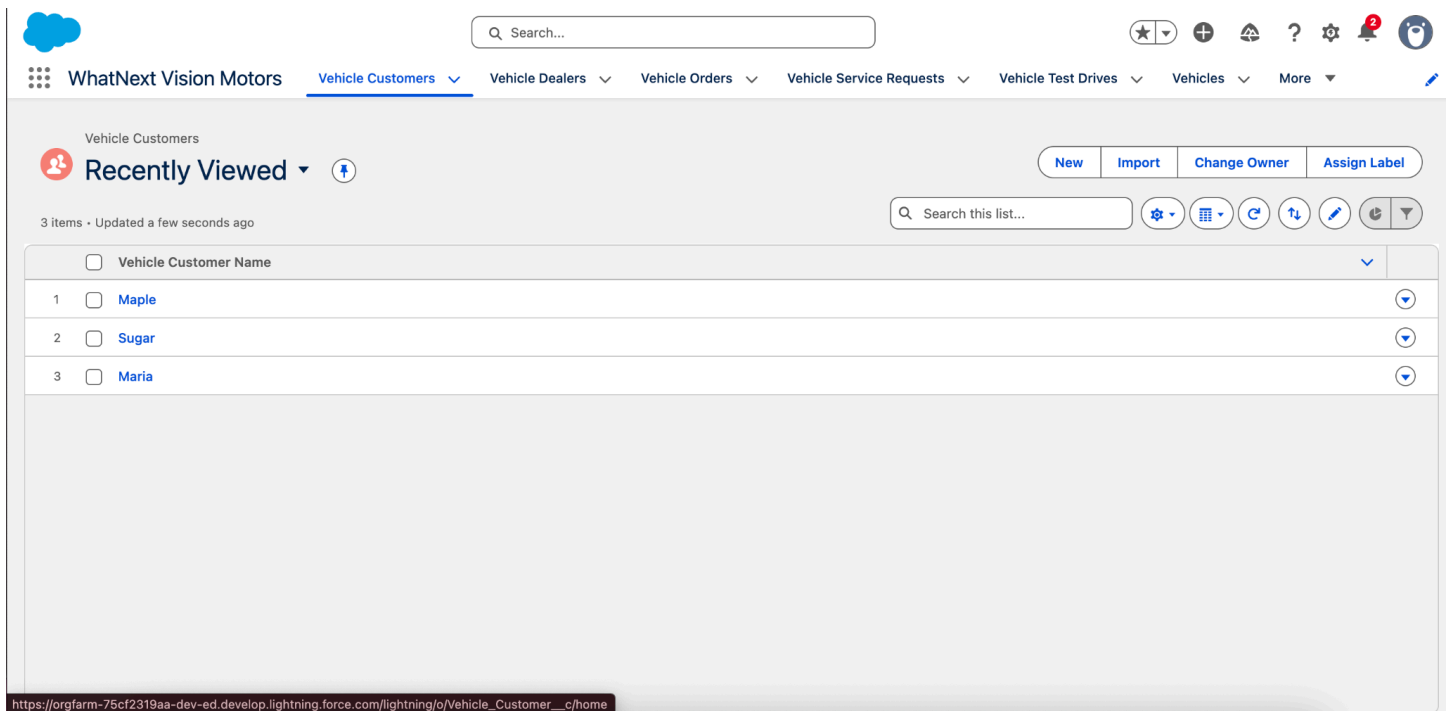
Logs Tests Checkpoints Query Editor View State Progress Problems

User	Application	Operation	Time	Status	Read	Size
------	-------------	-----------	------	--------	------	------

PHASE 3: UI/UX DEVELOPMENT & CUSTOMIZATION

→ Lightning App Setup

A dedicated “Vision Motors CRM” App was created with custom navigation items.



→ Page Layouts & Dynamic Forms

Layouts were customized for Sales, Dealer, and Vehicle records.



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WhatNext Vision Motors

Vehicle Customers

Vehicle Dealers

Vehicle Orders

Vehicle Service Requests

Vehicle Test Drives

Vehicles

More

Vehicle Orders

Recently Viewed

NewImportChange OwnerAssign Label

4 items • Updated a few seconds ago

	☐	Vehicle Order Number	
1	☐	O-0004001	
2	☐	O-0003001	
3	☐	O-0001001	
4	☐	O-0002001	

→Reports and Dashboards

Stock Summary Report



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WhatNext Vision Motors

Vehicle Customers

Vehicle Dealers

Vehicle Orders

Vehicle Service Requests

Vehicle Test Drives

Reports

More

Reports

All Reports

30 items

New ReportNew Folder

REPORTS	Report Name	Description	Folder	Created By	Created On	Subscribed
Recent	Exercise Completion by Days to Complete	Analyze an exercise's completion status based on average days to complete.	Enablement Dashboard Reports Summer '24	Automated Process	11/24/2025, 3:08 AM	
Created by Me	Exercise Completion by Days to Complete	Analyze an exercise's completion status based on average days to complete.	Enablement Dashboard Reports Spring '24	Automated Process	11/24/2025, 3:08 AM	
Private Reports	Exercise Completion by User	Analyze how long users are spending on an exercise and the amount of progress they're making.	Enablement Dashboard Reports Summer '24	Automated Process	11/24/2025, 3:08 AM	
Public Reports	Exercise Completion by User	Analyze how long users are spending on an exercise and the amount of progress they're making.	Enablement Dashboard Reports Spring '24	Automated Process	11/24/2025, 3:08 AM	
All Reports	Exercise Completion Status	Analyze how long users take to complete an exercise and their completion percentage.	Enablement Dashboard Reports Spring '24	Automated Process	11/24/2025, 3:08 AM	
FOLDERS	Exercise Completion Status	Analyze how long users take to complete an exercise and their completion percentage.	Enablement Dashboard Reports Summer '24	Automated Process	11/24/2025, 3:08 AM	
All Folders						
Created by Me						
Shared with Me						
FAVORITES						
All Favorites						

Orders by Status Dashboard

PHASE 4: DATA MIGRATION, TESTING & SECURITY

→ Data Migration

Initial customer and vehicle data were imported using Data Import Wizard.

→ Field History Tracking, Duplicate Rules & Matching Rules

Duplicate prevention rules were enabled for Customer and Dealer objects.

→ Profiles, Roles & Permission Sets

Role hierarchy ensured data visibility by business function.

→ Test Classes

Test classes were created to validate trigger behavior.

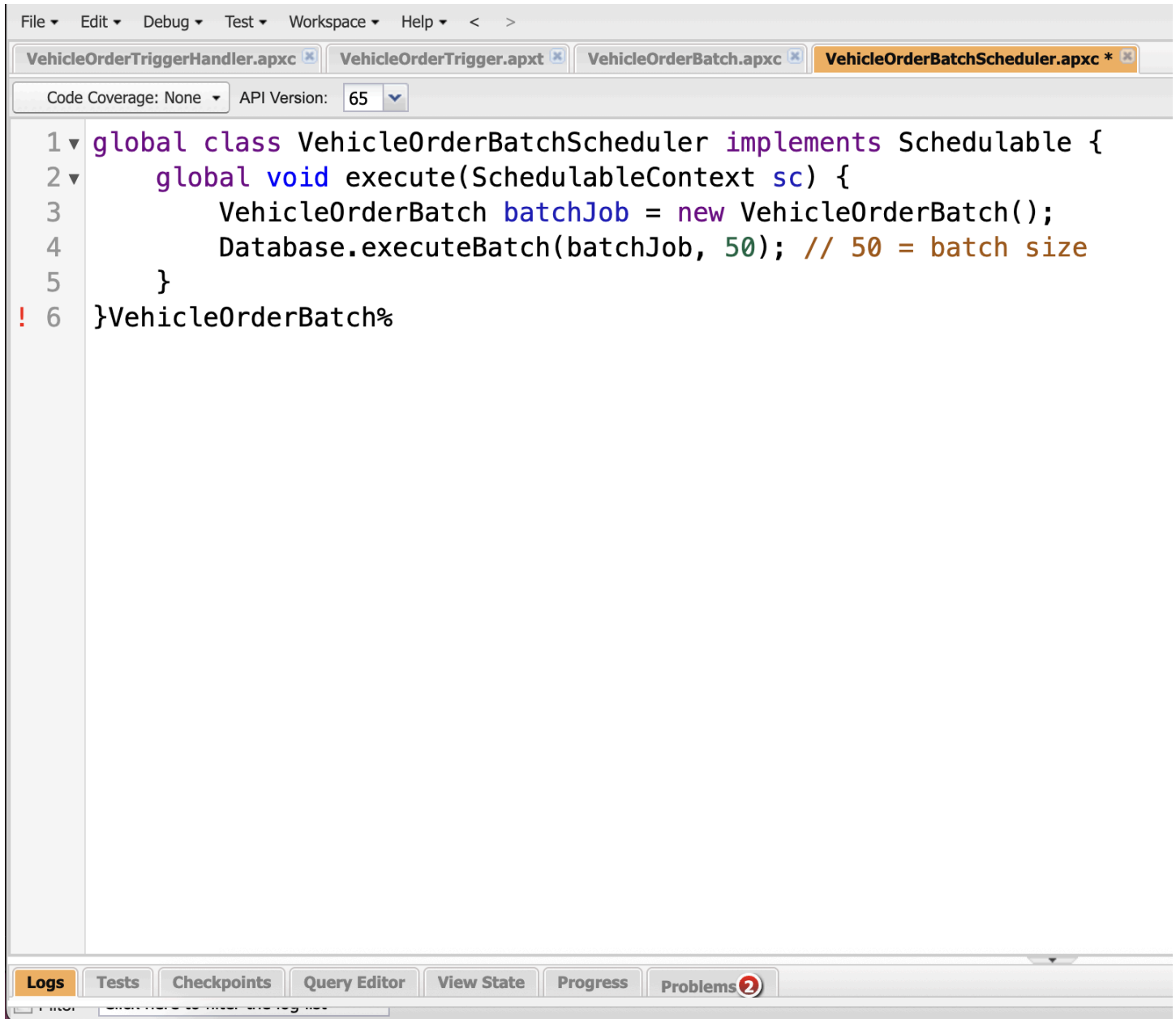
VehicleOrderBatch

```
File Edit Debug Test Workspace Help < >
VehicleOrderTriggerHandler.apxc VehicleOrderTrigger.apxt VehicleOrderBatch.apxc VehicleOrderBatchScheduler.apxc
Code Coverage: None API Version: 65
1 global class VehicleOrderBatch implements Database.Batchable<sObject> {
2
3     global Database.QueryLocator start(Database.BatchableContext bc) {
4         return Database.getQueryLocator([
5             SELECT Id, Status__c, Vehicle__c FROM Vehicle_Order__c WHERE Status__c = 'Pending'
6         ]);
7     }
8
9     global void execute(Database.BatchableContext bc, List<Vehicle_Order__c> orderList) {
10         Set<Id> vehicleIds = new Set<Id>();
11         for (Vehicle_Order__c order : orderList) {
12             if (order.Vehicle__c != null) {
13                 vehicleIds.add(order.Vehicle__c);
14             }
15         }
16
17         if (!vehicleIds.isEmpty()) {
18             Map<Id, Vehicle__c> vehicleStockMap = new Map<Id, Vehicle__c>([
19                 SELECT Id, Stock_Quantity__c FROM Vehicle__c WHERE Id IN :vehicleIds
20             ]);
21
22             List<Vehicle_Order__c> ordersToUpdate = new List<Vehicle_Order__c>();
23             List<Vehicle__c> vehiclesToUpdate = new List<Vehicle__c>();
24
25             for (Vehicle_Order__c order : orderList) {
26                 Vehicle__c vehicle = vehicleStockMap.get(order.Vehicle__c);
27                 if (vehicle != null && vehicle.Stock_Quantity__c > 0) {
28                     order.Status__c = 'Confirmed';
29                     vehicle.Stock_Quantity__c -= 1;
30                     ordersToUpdate.add(order);
31                     vehiclesToUpdate.add(vehicle);
32                 }
33             }
34
35             if (!ordersToUpdate.isEmpty()) update ordersToUpdate;
36             if (!vehiclesToUpdate.isEmpty()) update vehiclesToUpdate;
37         }
38     }
39 }
```

Logs Tests Checkpoints Query Editor View State Progress Problems

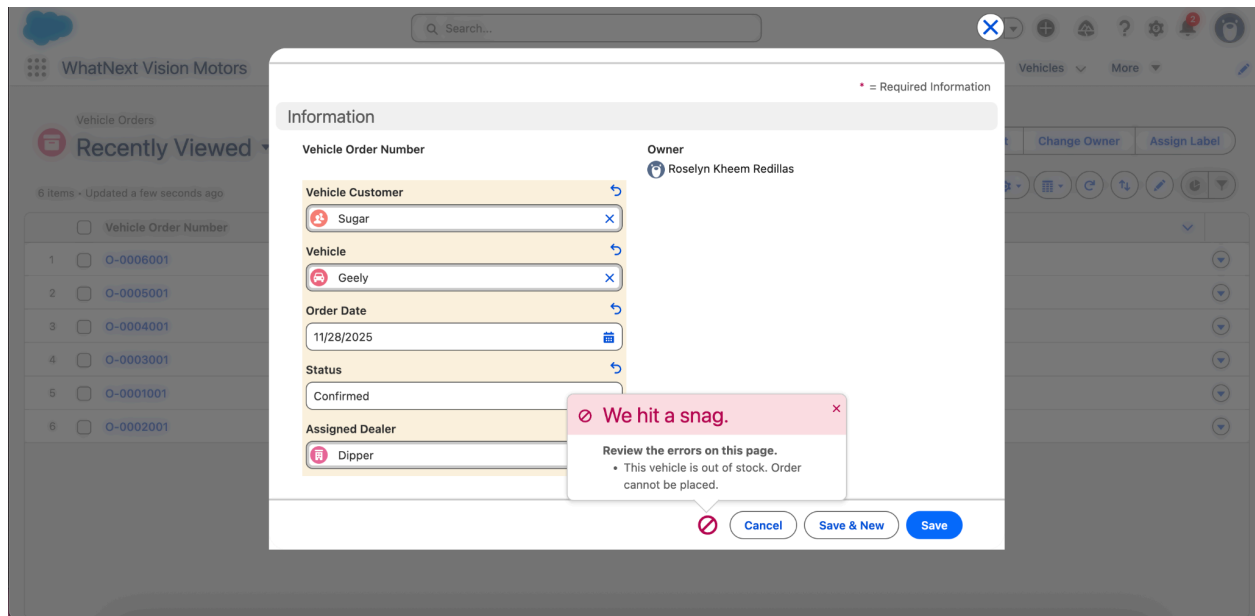
Filter Click here to filter the log list

VehicleOrderBatchScheduler

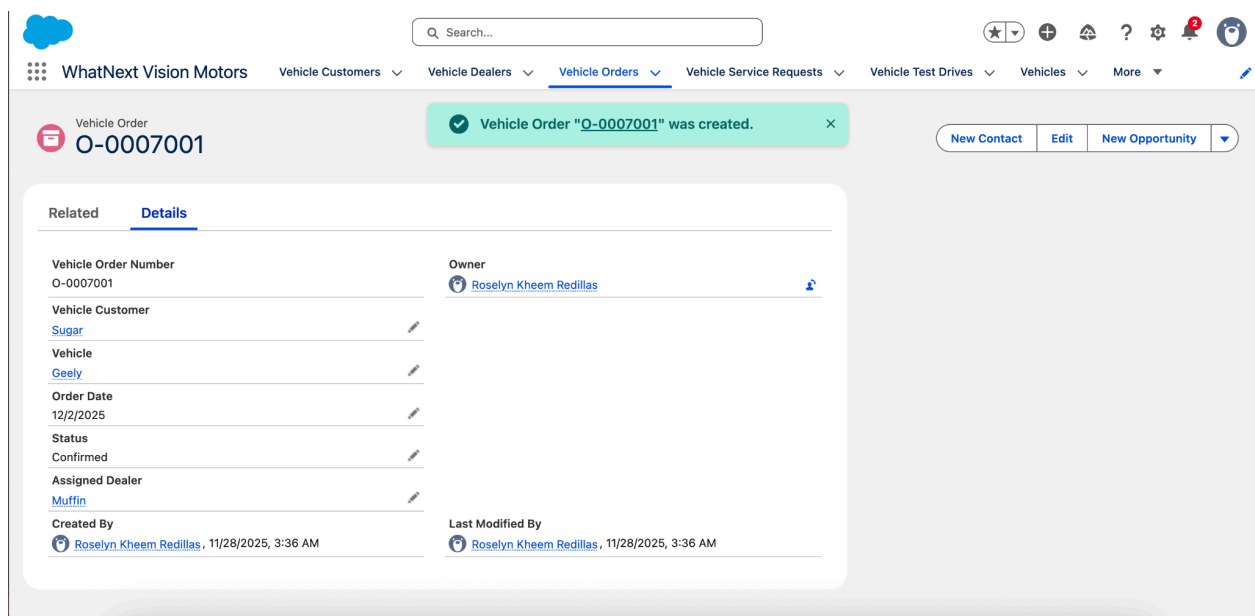


→Test Cases with Screenshots

- Creating an order with low stock → Expected Result: Fail



- Creating an order with sufficient stock → Expected Result: Success



PHASE 5: DEPLOYMENT, DOCUMENTATION & MAINTENANCE

→Deployment Strategy

Change Sets were used to deploy from Sandbox to Production.

→ **Maintenance & Monitoring**

The system will be monitored using debug logs, health checks, and regular stock updates.

→ **Troubleshooting Approach**

- Flow errors monitored via email alerts.
- Debug logs used for Apex troubleshooting.

CONCLUSION

The WhatNext Vision Motors CRM successfully enhances the customer lifecycle through automated dealer recommendations, real-time stock validation, and scheduled order status updates. This transformed ordering process increases transparency, reduces manual work, and improves customer satisfaction. The system is scalable, adaptable, and ready for future enhancements such as AI-driven recommendations and chatbot support.